

Video Decoder/Encoder Register Programming User Guide



Table of Contents

Introduction..... 2

Interpreting the Programming Script 2

 Settings Section..... 2

 Modes of Operation Sections..... 2

Using the Programming Script Evaluation Board Software 4



Introduction

This user guide contains information on how to interpret ADI's video products register programming script. The guide is targeted at users of an evaluation board system and users who require register programming writes for an ADI video product integrated in their own system.

The register programming script contains the recommended register writes for all standard modes that the product supports. The latest programming script for a video product is found on its product page at www.analog.com. The programming script can be found and downloaded from the Evaluation Board section of the product page. Note that the programming scripts for older generation products may not be available.

Interpreting the Programming Script

The programming script consists of I²C programming examples for standard modes of operation. These scripts are applicable to a system with the inputs arranged as shown in the Typical Connection diagram in the product datasheet. The input selection registers change in accordance with the PCB layout.

Settings Section

This section provides the revision number and release date of the script.

Modes of Operation Sections

The sections following the Settings section are register programming examples for the standard modes of operation section. Each section includes two headings that describe the mode of operation. It can also refer to the downstream data path of the video if an evaluation board is being used.

Example 1:

```
##SDP CVBS##  
:AUTODETECT NTSC/PAL/SECAM 10 Bit 422 Out through (Encoder in 525i mode):
```

This heading describes a mode of operation for CVBS input where the standard of NTSC, PAL, or SECAM can be auto-detected. The decoder will produce a 10-bit digital output in a 4:2:2 format. If using an evaluation board, the data will be passed to a video encoder programmed for 525i.

Video Decoder/Encoder Register Programming User Guide		
Rev. 0 March 29, 2006	Analog Devices B.V.	2



```

ADV7188_ADV7188@_ADV7321-VER.2.840.TXT - Notepad
File Edit Format View Help
*****

##SETTINGS##
:version 2.840 24th May 2005:
End

##SDP CVBS##
:AUTODETECT NTSC/PAL/SECAM 10 Bit 422 Out through (Encoder in 525i mode):
42 00 0E ; CVBS Ain10
42 03 00 ; 10 Bit Mode
42 04 57 ; Enable SFL
42 17 41 ; select SH1
42 1D 47 ; Enable 28MHz Crystal
42 19 FA ; split filter control
  
```

Figure 1. Extract from Programming Script

The recommended I²C programming writes directly follow the headings for each mode of operation.

Each line consists of three Hex values followed by a colon and some text:

- The first Hex value is the I²C device address. The device address for ADI decoders is 0x42 and 0x54 for encoders. Refer to the specific decoder/encoder datasheet for more detail. Any other lines in the programming script with an alternative device address are writes to other devices available on the Evaluation Board System.
- The second Hex value is the register address to which a new value must be written.
- The third Hex is the value that must be written to the given register address.

The text comment provides some information on why the write is necessary. Refer to the product datasheet for a description of the full register functionality.

Example 2:

42 00 0E ; CVBS Ain10

This line recommends programming a value of 0x0E to register address 0x00 on device address 0x42, which is the video decoder.



Using the Programming Script Evaluation Board Software

To enable easy and efficient evaluation, the ADV Register Control software is provided with decoder/encoder evaluation boards. The latest programming script for a video product may be found on its respective product page at www.analog.com and used in conjunction with the evaluation software. Note that older generation evaluation boards may not be provided with the ADV Register Control software.

Copy the programming script into the folder named Setup_Files in the ADV Register Control software files installed on your PC, that is, C:\Program Files\ADV Register Control 3.55\Setup_Files.

Do not change Setup_Files from the .txt format or change its file name, otherwise the software will not load correctly.

When the software is launched again, the option of the latest programming script will appear for selection in the first screen.

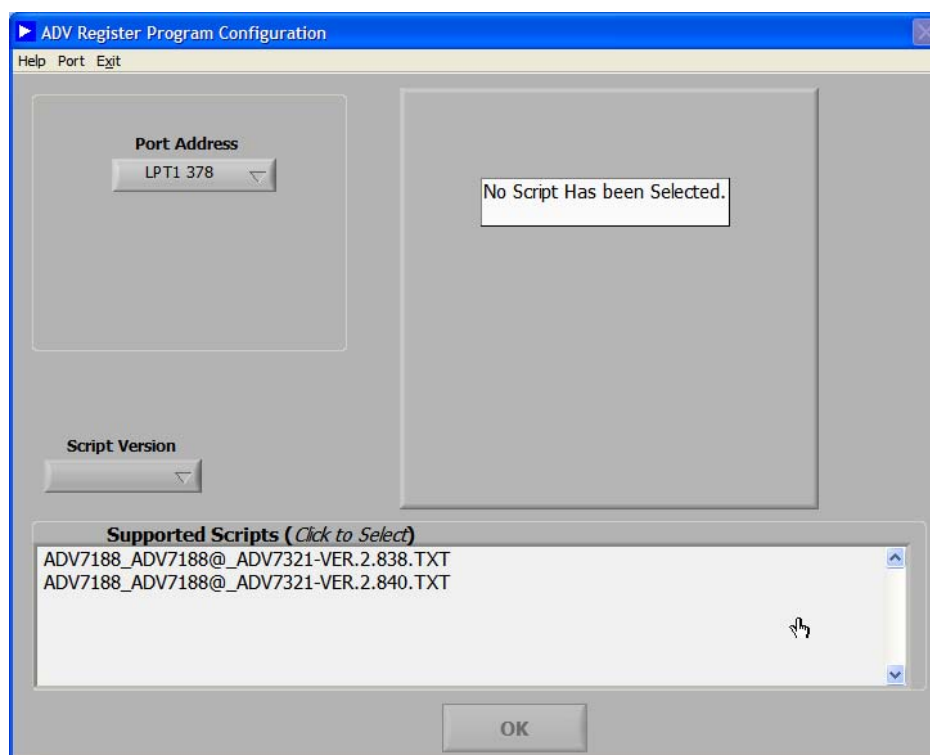


Figure 2. Programming Script Selection Screen